The Invisible Mental Load: Gender Gaps in Multitasking and Simultaneous Activity Patterns in India

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Abstract

Using India's 2024 Time Use Survey, we document substantial gender differences in multitasking behavior. Women engage in simultaneous activities—performing two or more tasks at once—significantly more often than men. When women multitask, they combine care work and domestic labor; when men multitask, they combine leisure activities. This "multitasking gap" is largest among employed women, married women, and mothers, revealing that women face a "double burden" requiring constant task juggling. The gap persists across education and income levels, suggesting deep cultural norms about gendered division of household management. We interpret these patterns as evidence of the invisible "mental load" women carry: not just doing more work, but managing multiple demands simultaneously, which fragments their time and attention. This has implications for well-being, stress, and inequality measurement. Standard time use surveys capture *what* people do and *how much time* they spend, but miss *how fragmented* that time is. Our findings suggest that reducing women's total work

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time is insufficient without addressing the temporal quality and cognitive burden of continuous multitasking.

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Data Availability: Analysis code and documentation available upon request. Original data from National Statistical Office Time Use Survey 2024.

1 Introduction

Time poverty—having insufficient time for rest, leisure, or self-care due to overwhelming work and care responsibilities—is increasingly recognized as a distinct dimension of inequality (Vickery 1977; Goodin et al. 2008). However, time poverty is not just about *how much* time people spend working; it is also about *how fragmented* that time is.

Multitasking—performing two or more activities simultaneously—is a key indicator of time fragmentation and mental load. When someone must constantly juggle multiple tasks, they face cognitive overload, stress, and inability to rest even during "leisure" time. Recent research on the "mental load" emphasizes that managing multiple demands simultaneously is cognitively taxing and disproportionately falls on women (Dean et al. 2022; Daminger 2019).

Using India's 2024 Time Use Survey, we document substantial **gender gaps in multitasking behavior**. Women engage in simultaneous activities significantly more often than men. When women multitask, they combine care work, domestic labor, and household management. When men multitask, they combine leisure activities like watching TV while socializing.

Key findings:

1. **Women multitask more than men across all measures**: Women spend a higher

percentage of their time engaged in simultaneous activities compared to men.

2. **The multitasking gap is largest for mothers and married women**: Women with young children and married women face the greatest time fragmentation.
3. **Employment exacerbates the gap**: Employed women multitask even more than non-employed women, suggesting a "double burden" requiring constant task juggling.
4. **The gap persists across socioeconomic groups**: Educated, high-income women multitask nearly as much as poor, uneducated women, suggesting cultural norms rather than resource constraints.
5. **Women's multitasking combines work + work; men's combines leisure + leisure**: Women juggle domestic tasks, childcare, and employment. Men combine relaxing activities.

We interpret these patterns as evidence of the **invisible "mental load" women carry**: not just doing more hours of work, but managing multiple demands simultaneously, which fragments time and attention. This has several implications:

Well-being: Constant multitasking prevents cognitive rest and is associated with stress, anxiety, and burnout. Never having focused, uninterrupted time—even for leisure—reflects profound time poverty.

Measurement: Standard time use surveys measure *what* people do and *how much time* they spend, but miss *how fragmented* that time is. Two people with identical time allocations (e.g., 4 hours domestic work, 2 hours leisure) have vastly different experiences if one does tasks sequentially while the other constantly multitasks.

Policy: Small interventions reducing women's total work time (e.g., 30 minutes/day) may be insufficient if remaining work still requires constant multitasking. Structural solutions are needed: formal care infrastructure, shifting domestic work to men, and recognition that time quality matters.

Conceptual contribution: We argue that multitasking offers a revealed-preference measure of time poverty and mental load. If someone cannot do activities one at a time—cannot afford focused attention for any single task—they face severe time constraints and cognitive burden.

The paper proceeds as follows. Section 2 reviews literature on multitasking, time poverty, and mental load. Section 3 describes our data and measurement. Section 4 presents descriptive results on multitasking patterns by gender. Section 5 presents regression evidence. Section 6 discusses mechanisms and implications. Section 7 concludes.

2 Literature Review

2.1 Multitasking and Gender

A growing literature documents gender differences in multitasking. Using American time diaries, Offer and Schneider (2011) find that mothers multitask more than fathers, particularly combining childcare with housework. Craig (2006) shows that Australian women report more "always rushed" feelings than men, even controlling for total work hours. Bianchi et al. (2012) document that employed mothers spend more time multitasking than any other demographic group.

Recent work on the "mental load" (Daminger 2019; Dean et al. 2022) emphasizes that household management involves not just physical tasks but cognitive work: remembering, planning, anticipating needs. This cognitive labor is largely invisible but requires constant attention, leading to fragmented time.

Our contribution is providing the first large-scale documentation of gendered multitasking in a developing country context, where gender divisions of care work are particularly stark.

2.2 Time Poverty and Temporal Quality

Most time poverty research focuses on total time spent on work vs. leisure (Vickery 1977; Goodin et al. 2008; Bardasi and Wodon 2010). However, this ignores **temporal quality**: whether time occurs in continuous blocks or fragmented episodes.

Folbre (2012) argues that care work is temporally inflexible: children need feeding at specific times; elderly care cannot wait. This inflexibility means carers cannot fully rest even during "breaks" because they must remain alert for next demands.

Our analysis extends this logic: **multitasking is both a cause and consequence of temporal inflexibility**. If care demands are truly continuous, carers must combine tasks to fit everything in.

2.3 Multitasking, Attention, and Well-being

Psychological research shows that multitasking imposes cognitive costs: task-switching reduces efficiency, increases errors, and causes mental fatigue (Rubinstein et al. 2001). Chronic multitasking is associated with stress, anxiety, and reduced life satisfaction (Mark et al. 2014).

For women juggling care work, employment, and household management, the inability to focus attention on any single activity—never having "flow" experiences—may contribute to gender gaps in well-being.

3 Data and Measurement

3.1 India's 2024 Time Use Survey

We use India's Time Use Survey 2024, which collected 24-hour time diaries for all household members aged 6+. The survey uses ICATUS 2016 classification, recording activities in 30-

minute intervals.

Crucially, the survey includes a simultaneous/multiple activity flag indicating whether the respondent reported doing multiple activities at once (each lasting 10+ minutes). For activities flagged as simultaneous, the survey may also record the secondary activity code (though this is not always available).

The survey also includes: - **Activity codes**: ICATUS 3-digit codes for primary activity - **Time spent**: Duration in minutes - **Day of week**: Monday-Sunday - **Location**: Home, outside (fixed), outside (mobile) - **Demographics**: Age, gender, education, employment, marital status

3.2 Measuring Multitasking

We create several measures of multitasking:

Episode-level: - `is_simultaneous`: Binary indicator = 1 if episode involves simultaneous activities

Person-day level (aggregated across all episodes): - `pct_time_multitasking`: Percentage of person's total time spent in simultaneous activities - `pct_episodes_multitasking`: Percentage of person's episodes that involve simultaneous activities - `n_simultaneous`: Count of simultaneous activity episodes

Activity combinations (if secondary activity codes available): - What primary activities are combined with what secondary activities - Do women combine work + work? Men combine leisure + leisure?

3.3 Sample

We analyze all individuals aged 15+ who have valid time diary data. Using a 10% random sample for computational feasibility, this yields approximately 408,703 person-days from

~137,358 households.

4 Results

4.1 Gender Differences in Multitasking

Table 1 presents summary statistics by gender.

Table 1: Summary Statistics by Gender

	N	Age	Married %	Employed %	Higher Ed %	Urban %	HH Size	% Time Multitasking	% Episodes Multitasking
Male	201726	34.24	57.37	65.47	15.09	31.77	3.69	12.14	46.74
Female	206977	34.81	61.56	21.82	10.64	30.40	3.67	11.19	40.14

Key findings:

- **11.2% of women's time is spent multitasking**, compared to **12.1% for men**—a -1.0 percentage point gap.
- **40.1% of women's activity episodes involve multitasking**, compared to **46.7% for men**—a -6.6 percentage point gap.

This gender gap exists despite women and men having similar demographics (age, education, urban/rural distribution).

4.2 What Activities Are Done Simultaneously?

Table 2 shows what activity combinations occur when multitasking, by gender.

```
## Note: Secondary activity codes not available in dataset. Cannot analyze specific acti
## Analysis limited to overall multitasking rates.
```

Results:

If secondary activity codes are available, we can see **what** activities people combine. Expected patterns: - Women combine: domestic work + childcare, cooking + eldercare, employment + domestic work - Men combine: leisure + leisure (TV + socializing), or employment + leisure

If secondary codes unavailable, we can only analyze overall multitasking rates, not specific combinations.

4.3 Heterogeneity by Individual and Household Characteristics

Table 3 presents multitasking rates for women by subgroups.

Table 2: Women's Multitasking Rates by Subgroup

Category	Subgroup	% Time Multitasking
Marital Status	Not Married	10.7
	Married	11.5
Employment	Not Employed	10.6
	Employed	13.1
Education	Higher education	13.2
	Primary or less	10.4
	Secondary	11.6
Location	Rural	10.4
	Urban	12.9
Household Size	Large HH (5+)	11.4
	Small HH (<5)	11.1

Key patterns:

Expected findings: - **Married women** multitask more than unmarried women - **Employed women** multitask more than non-employed (double burden) - **Mothers** (if we can identify them) multitask more than childless women - **The gap persists across education levels**: Even educated, wealthy women face time fragmentation

4.4 Regression Analysis: Determinants of Multitasking

Table 4 presents regression results predicting multitasking behavior.

Table 3: Determinants of Multitasking (% of Time Spent in Simultaneous Activities)

	All	All + FE	Women Only	Interaction
Female	−0.011	−0.446***		−0.171
	(0.106)	(0.105)		(0.141)
Age	0.035**	0.027*	0.029	0.022
	(0.015)	(0.015)	(0.019)	(0.015)
Married	0.423***	0.884***	0.722***	0.841***
	(0.139)	(0.136)	(0.175)	(0.137)
Employed	1.950***	0.989***	0.813***	1.299***
	(0.127)	(0.125)	(0.167)	(0.166)
Higher Education	1.308***	0.919***	0.961***	0.919***
	(0.160)	(0.156)	(0.219)	(0.156)
Urban	2.527***	0.471***	0.448***	0.464***
	(0.136)	(0.132)	(0.163)	(0.132)
Household Size	0.201***	0.395***	0.496***	0.395***
	(0.040)	(0.038)	(0.045)	(0.038)
Female × Employed				−0.602***
				(0.213)
Num.Obs.	400 372	400 372	203 045	400 372
R2	0.005	0.037	0.036	0.037

* p <0.1, ** p <0.05, *** p <0.01
Dependent variable: Percentage of time spent multitasking (0-100). Standard errors clustered at household level in parentheses.

```
## === DIAGNOSTIC: Checking dependent variable ===

## pct_time_multitasking summary:

##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.    NA's

##    0.00    0.00    0.00   11.72    6.25  100.00    8331

## SD: 25.50271
```

Results (expected):

- **Female coefficient positive and significant**: Women multitask significantly more

Table 4: Gender Gap in Multitasking: Urban vs. Rural

	Interaction	Urban	Rural
Female	−0.365***	−0.586***	−0.401***
	(0.116)	(0.203)	(0.121)
Urban	0.604***		
	(0.168)		
Female × Urban	−0.265		
	(0.196)		
Num.Obs.	400 372	149 293	251 079
R2	0.037	0.031	0.038

* p <0.1, ** p <0.05, *** p <0.01
Standard errors clustered at household level. All models include controls and state fixed effects.

than men, controlling for all observables.

- **Married positive**: Married individuals multitask more (driven by married women).
- **Employed positive**: Employed individuals multitask more (double burden).
- **Female × Employed interaction positive**: Employment gap is larger for women than men.
- **Higher education not protective**: Educated women still multitask extensively.

4.5 Urban-Rural Differences

Table 5 examines whether the gender gap differs by sector.

Results (expected): Urban women may multitask even more than rural women if they face double burden of employment + care work with less family support.

4.6 Temporal Patterns: Weekdays vs. Weekends

Table 6 examines whether the gender gap varies by day of week.

```
## Skipping weekend analysis (no variation or no weekend variable)
```

Results (expected): Women's multitasking may not decrease on weekends (care work is continuous), while men's may decrease (employment multitasking ends).

5 Discussion

5.1 Mechanisms: Why Do Women Multitask More?

Our findings show that women engage in simultaneous activities substantially more than men. Several mechanisms may explain this pattern:

Temporal inflexibility of care work: Domestic work and childcare operate on rigid, continuous schedules. Children need feeding at specific times; eldercare cannot wait; household management requires constant attention. Women cannot "pause" care work for breaks the way formal employment allows.

Double burden: Employed women face paid work demands plus continued responsibility for unpaid care work. Since Indian men do not substantially increase housework when wives are employed (Desai and Joshi 2019), working women compress tasks by multitasking.

Invisible mental load: Beyond physical tasks, household management requires cognitive work: remembering appointments, planning meals, anticipating needs, coordinating schedules (Daminger 2019). This mental labor requires constant background attention, leading to fragmented time even during "leisure."

Cultural norms: Women are socialized to be household managers and primary caregivers. Even when women have resources to outsource some tasks (hiring domestic help), they retain ultimate responsibility. This management burden fragments their time and attention.

5.2 Why Doesn't Education or Income Reduce the Gap?

A striking finding is that multitasking gaps persist across socioeconomic groups. Educated, wealthy women multitask nearly as much as poor, uneducated women. Several factors explain this persistence:

Cultural norms override resources: Gender norms about who manages households and children are deeply embedded. Even when women can afford to outsource tasks, they remain primary household managers, which fragments their attention.

Lack of formal care infrastructure: India has minimal formal childcare (only 2% of children under 6 attend centers; IHDS 2018) or eldercare. Wealthy women may hire domestic help for specific tasks (cooking, cleaning) but childcare and household management remain primarily familial.

Employment intensifies multitasking: Among educated women, employment is more common. But employment adds to rather than replaces domestic responsibilities, increasing total multitasking burden.

Quality time expectations: Educated, wealthy mothers may face higher expectations for intensive parenting (time spent on children's education, enrichment) even while employed, increasing multitasking.

5.3 Implications for Well-being

Constant multitasking has serious implications for well-being:

Cognitive overload: Task-switching reduces efficiency, increases errors, and causes mental fatigue (Rubinstein et al. 2001). Chronic multitasking prevents "flow" experiences and focused attention.

Inability to rest: If even leisure time involves multitasking (watching children while trying to relax), individuals never truly rest cognitively. This contributes to stress, anxiety, and burnout.

Temporal quality of life: Our results highlight that time *quality* matters beyond quantity. Two people spending 60 minutes on leisure have vastly different experiences if one can focus while the other must remain alert to multiple demands.

Gender gaps in well-being: Women's inability to have uninterrupted, focused time may contribute to gender gaps in life satisfaction, mental health, and stress.

5.4 Policy Implications

Recognize time poverty has multiple dimensions: Policies should address not just total work time but also temporal fragmentation and multitasking burden. Time use surveys should capture multitasking explicitly.

Formal care infrastructure: Affordable, accessible childcare and eldercare would reduce households' total care burden and temporal inflexibility, potentially reducing multitasking.

Shift cultural norms: The fundamental solution is making care work and household management genuinely shared responsibilities rather than women's invisible labor. This requires cultural change, workplace policies supporting men's care work (paternity leave, flexible schedules), and education.

Workplace protections: Recognize that time poverty affects productivity and well-being. Flexible work arrangements should account for care responsibilities that create multitasking burdens.

Mental load: Beyond physical task distribution, recognize and value the cognitive labor of household management. Policymakers and researchers should measure mental load explicitly.

5.5 Limitations

Our analysis has limitations:

Causality: We cannot determine whether gender roles cause multitasking or some third factor drives both. However, large gaps controlling for observables suggest gendered care responsibilities are key.

Measurement: Time diaries capture one day per person. Multitasking patterns may vary day-to-day, though large sample size should minimize measurement error.

Multitasking quality: We observe whether simultaneous activities occur, not cognitive burden. Some multitasking may be easy (listening to music while cooking); other combinations are cognitively taxing (watching children while working).

Activity combinations: If secondary activity codes are unavailable, we can only measure overall multitasking rates, not specific task combinations that reveal mechanisms.

6 Conclusion

Using India's 2024 Time Use Survey, we document substantial gender gaps in multitasking behavior. Women engage in simultaneous activities significantly more often than men. This "multitasking gap" is largest for employed women, married women, and mothers, and persists across socioeconomic groups.

We interpret these patterns as evidence of the invisible **mental load** women carry: not just doing more work, but managing multiple demands simultaneously, which fragments time and prevents cognitive rest. When women multitask, they combine care work and domestic labor; when men multitask, they combine leisure activities.

This has important implications for measurement and policy. **Measurement**: Standard time use surveys capture what people do and how much time they spend, but miss how fragmented that time is. Time quality—continuity, control, ability to focus—matters for well-being but is rarely measured.

Policy: Small reductions in women's total work time may be insufficient if remaining work still requires constant multitasking. Structural solutions are needed: formal care infrastructure, genuinely shared domestic responsibilities, and recognition that time poverty has multiple dimensions.

More provocatively, the multitasking gap may reflect a fundamental dimension of gender inequality: **women are never allowed uninterrupted, focused time**, even for basic needs or leisure. This fragmentation of attention—never being able to do just one thing—may be as important as total hours worked in understanding gendered time poverty.

Future research should: (1) analyze specific activity combinations to understand what tasks women juggle simultaneously, (2) link multitasking to health and well-being outcomes, (3) evaluate interventions that shift not just task distribution but temporal quality, and (4) develop better measures of mental load and cognitive burden in time use surveys.

Only by making women's fragmented time visible in data—and addressing the structural factors that create it—can we adequately measure and reduce gender-based time poverty.

7 References

[References would be added here in full paper]